

Key features of parabolas



1 Consider the general quadratic equation $y = ax^2 + bx + c$, $a \neq 0$.

a If $a < 0$, in what direction will the parabola open?

b If $a > 0$, in what direction will the parabola open?

2 Does the parabola represented by the equation $y = x^2 - 8x + 9$ open upward or downward?

3 Does the graph of $y = x^2 + 6$ have any x -intercepts? Explain your answer.

4 State whether the following parabolas have x -intercepts:

a $y = (x - 7)^2 + 4$

b $y = -(x - 7)^2 + 4$

c $y = -(x - 7)^2 - 4$

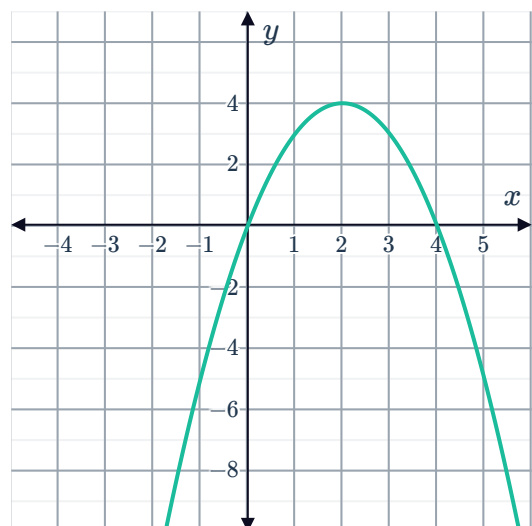
d $y = (x - 7)^2 - 4$

5 Consider the given graph:

a What are the x -intercepts?

b What is the y -intercept?

c What is the maximum value?



6 Consider the given graph:

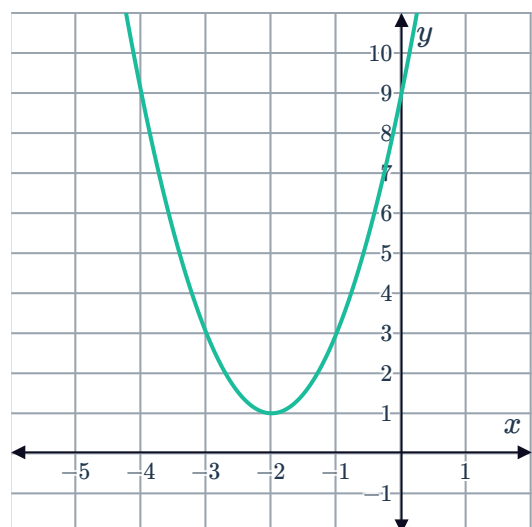
a Is the curve concave up or concave down?

b State the y -intercept of the graph.

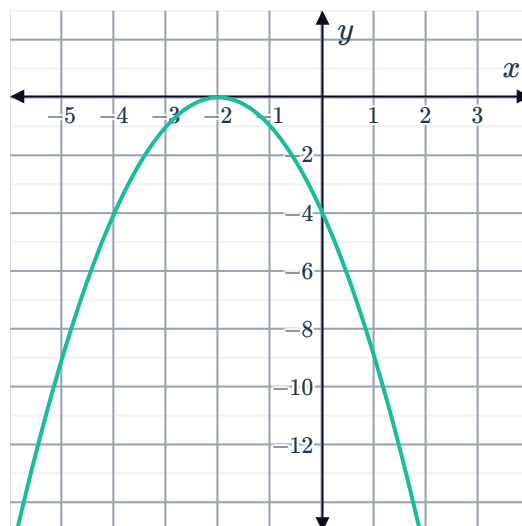
c What is the minimum value?

d At which value of x does the minimum value occur?

e Determine the interval of x for which the graph is decreasing.



7 Consider the graph of the parabola:



- a State the coordinates of the x -intercept.
- b State the coordinates of the vertex.
- c State whether the following statements are true about the vertex:
 - i The vertex is the minimum value of the graph.
 - ii The vertex occurs at the x -intercept.
 - iii The vertex lies on the axis of symmetry.
 - iv The vertex is the maximum value of the graph.

8 Suppose that a particular parabola is concave down, and its vertex is located in quadrant 2.

- a How many x -intercepts will the parabola have?
- b How many y -intercepts will the parabola have?

9 Suppose that a particular parabola has two x -intercepts, and its vertex is located in quadrant 4. Will such a parabola be concave up or concave down?

10 Consider the quadratic function defined in the table on the right:

- a What are the coordinates of the vertex?
- b What is the minimum value of the function?

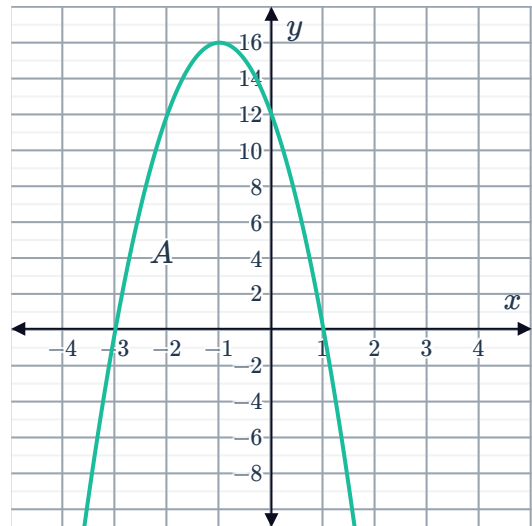
x	y
-7	11
-6	6
-5	3
-4	2
-3	3
-2	6
-1	11

11 A vertical parabola has an x -intercept at $(-1, 0)$ and a vertex at $(1, -6)$. Find the other x -intercept.

12 State whether the following can be found, without any calculation, from the equation of the form $y = (x - h)^2 + k$ but not from the equation of the form $y = x^2 + bx + c$:



- 13 Quadratic function A is represented graphically as shown. Quadratic function B , which is concave down, shares the same x -intercepts as quadratic function A , but has a y -intercept closer to the origin. Which of the functions has a greater maximum value?



- 14 What is the axis of symmetry of the parabola $y = k(x - 7)(x + 7)$ for any value of k ?
- 15 Consider the equation $y = 25 - (x + 2)^2$. What is the maximum value of y ?
- 16 Consider the function $y = (14 - x)(x - 6)$.
- a State the zeros of the function.
 - b Find the axis of symmetry.
 - c Is the graph of the function concave up or concave down?
 - d Determine the maximum y -value of the function.
- 17 Consider the parabola of the form $y = ax^2 + bx + c$, where $a \neq 0$.

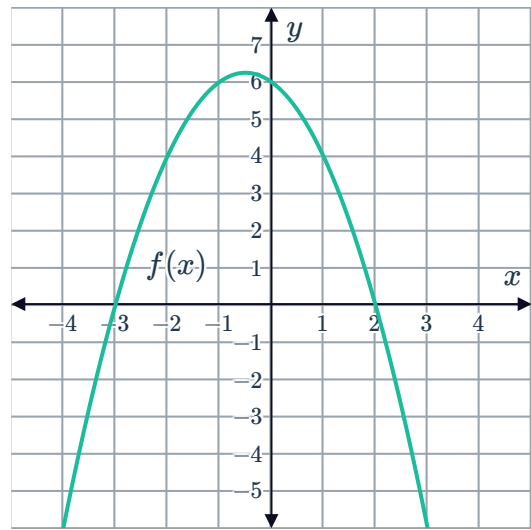
Complete the following statement:

The x -coordinate of the vertex of the parabola occurs at $x = \boxed{}$. The y -coordinate of the vertex is found by substituting $x = \boxed{}$ into the parabola's equation and evaluating the function at this value of x .

18 Consider the graph of the function

$$f(x) = -x^2 - x + 6:$$

Using the graph, write down the solutions of the equation $-x^2 - x + 6 = 0$.



19 True or false:

- a The quadratic formula can be used to find the y -intercept.
- b If the parabola has only one x -intercept, then the x -intercept is also the vertex.

20 Consider the parabola whose equation is $y = 3x^2 + 3x - 7$. Find the x -intercepts of the parabola in exact form.

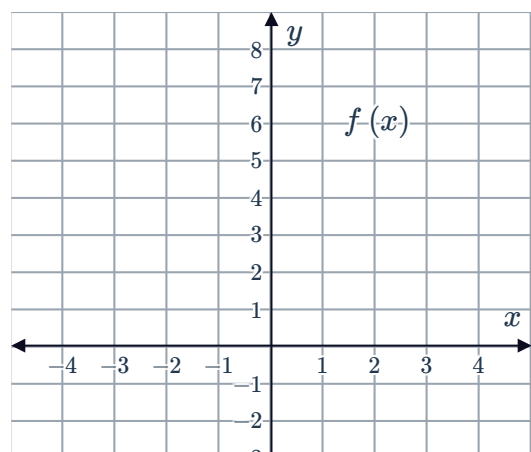
Graphing quadratics

21 Consider the parabola described by the function $y = -2x^2 + 2$.

- a Is the parabola concave up or down?
- b Is the parabola more or less steep than the parabola $y = x^2$?
- c What are the coordinates of the vertex of the parabola?
- d Sketch the graph of $y = -2x^2 + 2$.

22 Consider the two graphs. One of them has equation $f(x) = -x^2 + 5$.

- a What is the equation of the other graph?
- b Use the graph to approximate the solutions of the equation $-x^2 + 5 = 0$ to one decimal place.





23 Consider the quadratic function $h(x) = x^2 + 2$.

- a Sketch the graph of the parabola $h(x)$.
- b Plot the axis of symmetry of the parabola on the same graph.
- c What is the vertex of the parabola?

 **24** Consider the equation $y = (x - 3)^2 - 1$.

- a Find the x -intercepts.
- b Find the y -intercept.
- c Determine the coordinates of the vertex.
- d Sketch the graph.
- e Use your graph to find the solutions of the equation $(x - 3)^2 - 1 = 0$.

 **25** Consider the quadratic function $f(x) = -3(x + 2)^2 - 4$.

- a What are the coordinates of the vertex of this parabola?
- b What is the equation of the axis of symmetry of this parabola?
- c What is the y -coordinate of the graph of $f(x)$ at $x = -1$?
- d Sketch the graph of the parabola.
- e Plot the axis of symmetry of the parabola on the same graph.

26 On a number plane, sketch the shape of a parabola of the form $y = a(x - h)^2 + k$ that has the following signs for a , h and k :

- | | |
|-------------------------|-------------------------|
| a $a > 0, h > 0, k > 0$ | b $a < 0, h > 0, k > 0$ |
| c $a > 0, h > 0, k < 0$ | d $a < 0, h > 0, k < 0$ |



27 Consider the parabola $y = (2 - x)(x + 4)$.

- a State the y -intercept.
- b State the x -intercepts.
- c Complete the table of values:
- d Determine the coordinates of the vertex of the parabola.
- e Sketch the graph of the parabola.

x	-5	-3	-1	1	3
y					

28 Consider the parabola $y = (x - 3)(x - 1)$.

- a Find the y -intercept.
- b Find the x -intercepts.
- c State the equation of the axis of symmetry.
- d Find the coordinates of the turning point.
- e Sketch the graph of the parabola.

29 Consider the parabola $y = x(x + 6)$.

- a Find the y -intercept.
- b Find the x -intercepts.
- c State the equation of the axis of symmetry.
- d Find the coordinates of the turning point.
- e Sketch the graph of the parabola.

30 Sketch the graph of the following:

- a $y = (x + 2)(x - 3)$
- b $y = (x - 3)(x + 1)$

31 Consider the function $y = (x + 5)(x + 1)$.

- a Sketch the graph.
- b Sketch the graph of $y = -(x + 5)(x + 1)$ on the same set of axes.

32 Consider the equation $y = x^2 - 6x + 8$.

- a Factorise the expression $x^2 - 6x + 8$.
- b Hence, or otherwise, find the x -intercepts of the quadratic function $y = x^2 - 6x + 8$
- c Find the coordinates of the turning point.

d Sketch the graph of the function.



 **33** Consider the parabola $y = x^2 + x - 12$.

- a Find the x -intercepts of the curve.
- b Find the y -intercept of the curve.
- c What is the equation of the vertical axis of symmetry for the parabola?
- d Find the coordinates of the vertex of the parabola.
- e Sketch the graph of $y = x^2 + x - 12$.

 **34** A parabola has the equation $y = x^2 + 4x - 1$.

- a Express the equation of the parabola in the form $y = (x - h)^2 + k$ by completing the square.
- b Find the y -intercept of the parabola.
- c Find the vertex of the parabola.
- d Is the parabola concave up or down?
- e Hence, sketch the graph of $y = x^2 + 4x - 1$.
- f Use your graph to approximate the solutions of the equation $x^2 + 4x - 1 = 0$.

 **35** Consider the quadratic $y = x^2 - 12x + 32$.

- a Find the zeros of the quadratic function.
- b Express the equation in the form $y = a(x - h)^2 + k$ by completing the square.
- c Find the coordinates of the vertex of the parabola.
- d Hence, sketch the graph.

 **36** Consider the curve $y = x^2 + 6x + 4$.

- a Determine the axis of symmetry.
- b Hence, determine the minimum value of y .
- c Sketch the graph of the function.
- d Use your graph to approximate the solutions of the equation $x^2 + 6x + 4 = 0$.

 **37** Consider the function $P(x) = -2x^2 - 8x + 2$.

- a Find the coordinates of the vertex.
- b Sketch the graph.
- c Use your graph to approximate the solutions of the equation $-2x^2 - 8x + 2 = 0$.



 **38** Consider the equation $y = 6x - x^2$.



- a** Find the x -intercepts of the quadratic function.
- b** Find the coordinates of the turning point.
- c** Sketch the graph.

 **39** A parabola is described by the function $y = 2x^2 + 9x + 9$.

- a** Find the x -intercepts of the parabola.
- b** Find the y -intercept for this curve.
- c** Find the axis of symmetry.
- d** Find the y -coordinate of the vertex of the parabola.
- e** Sketch the graph.

Graphing quadratics using technology

40 Use your calculator or other handheld technology to graph the equations below. Then answer the following questions:

- | | |
|---|---------------------------------------|
| i What is the vertex of the graph? | ii What is the y -intercept? |
| a $y = 4x^2 - 64x + 263$ | b $y = -4x^2 - 48x - 140$ |

41 Use your calculator or other handheld technology to graph $y = -3x^2 - 12$.

- a** What is the vertex of the graph?
- b** Are there any x -intercepts?
- c** For what values of x is the parabola decreasing?

42 Use technology to graph the parabola $y = -2x^2 + 16x - 24$.

- a** Find the x -intercepts of the parabola.
- b** Find the y -intercept of the parabola.
- c** Find the axis of symmetry of the parabola.
- d** Find the y -coordinate of the vertex of the parabola.